

ECA5612 –Advanced Wireless Communication Systems 5G / 6G Communication Systems.

Experiment 15 : AI-based Traffic Prediction for 5G Networks

Objective 1: To simulate 5G network traffic data and visualize the traffic pattern.

Program:

```
clc;

clear;

close all;

% Simulated 5G Network Traffic (Mbps)

time = 1:20;

traffic = [20 25 30 35 40 50 60 58 65 70 ...
          75 80 78 85 90 95 100 98 105 110];

figure;

plot(time, traffic, '-o', 'LineWidth', 2);

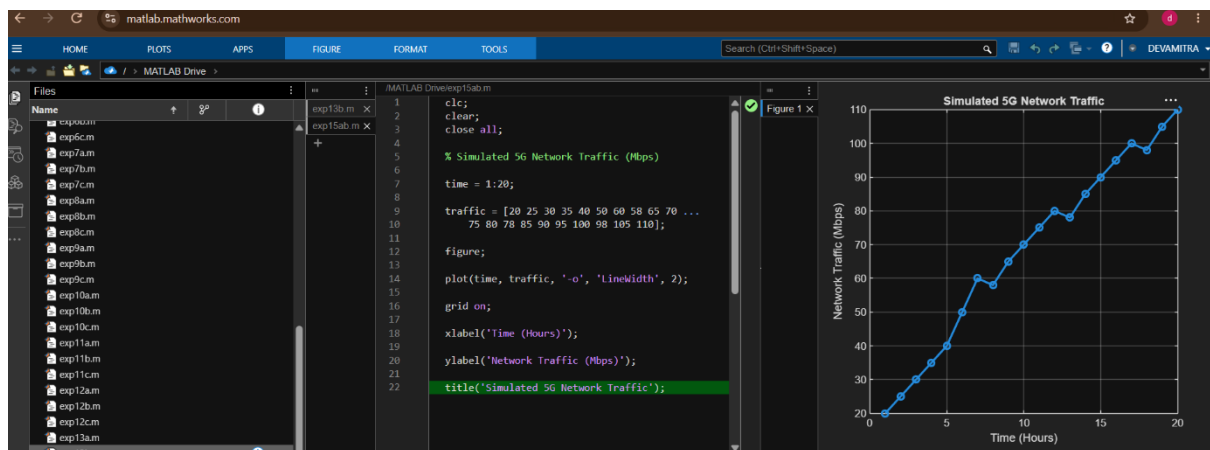
grid on;

xlabel('Time (Hours)');

ylabel('Network Traffic (Mbps)');

title('Simulated 5G Network Traffic');
```

Output:



Objective 2: To predict future 5G network traffic using Machine Learning (Linear Regression).

Program:

```
clc;

clear;

close all;

% Simulated Traffic Data

time = 1:20;

traffic = [20 25 30 35 40 50 60 58 65 70 ...

          75 80 78 85 90 95 100 98 105 110];

% Linear Regression Model

model = polyfit(time, traffic, 1);

% Predicted Traffic

predicted = polyval(model, time);

figure;

plot(time, traffic, 'bo-', 'LineWidth', 2);

hold on;

plot(time, predicted, 'r*- ', 'LineWidth', 2);

grid on;

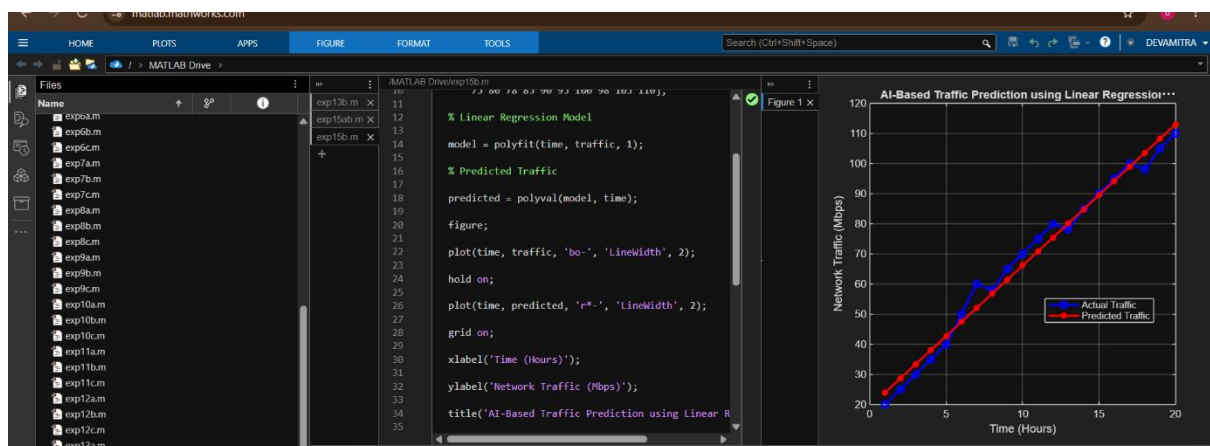
xlabel('Time (Hours)');

ylabel('Network Traffic (Mbps)');

title('AI-Based Traffic Prediction using Linear Regression');

legend('Actual Traffic', 'Predicted Traffic', 'Location', 'best');
```

Output:



Objective 3: To evaluate the prediction accuracy by comparing actual and predicted traffic values.

Program:

```
clc;

clear;

close all;

% Simulated Traffic Data

time = 1:20;

traffic = [20 25 30 35 40 50 60 58 65 70 ...
          75 80 78 85 90 95 100 98 105 110];

% Linear Regression Model

model = polyfit(time, traffic, 1);

% Predicted Traffic

predicted = polyval(model, time);

% Calculate Prediction Error

error = traffic - predicted;

% Calculate RMSE

RMSE = sqrt(mean(error.^2));

disp(['Root Mean Square Error (RMSE) = ', num2str(RMSE)]);

figure;

% Graph 1: Actual vs Predicted Traffic

subplot(2,1,1);

plot(time, traffic, 'bo-', 'LineWidth', 2);

hold on;

plot(time, predicted, 'r*-', 'LineWidth', 2);

grid on;

xlabel('Time (Hours)');

ylabel('Traffic (Mbps)');

title('Actual vs Predicted Traffic');

legend('Actual', 'Predicted', 'Location', 'best');

% Graph 2: Prediction Error

subplot(2,1,2);
```

```
bar(time, error);
```

```
grid on;
```

```
xlabel('Time (Hours)');
```

```
ylabel('Prediction Error (Mbps)');
```

```
title('Prediction Error');
```

```
% Display Results
```

```
disp('Time Actual Predicted');
```

```
disp([time' traffic' predicted']);
```

Output:

